Notebook 8.1 - Multi-Objective Optimization

Management Science - Finding the Best Trade-offs



Introduction

Welcome back to Bean Counter! As CEO, you're facing a critical decision: which new coffee product should we add to our menu in more than thousand locations?

Unlike previous decisions where we optimized a single metric (like minimizing delivery distance), today we'll learn to balance **multiple competing objectives** simultaneously. Specifically, we want to:

- **Maximize profit margin** (more revenue per cup)
- **Minimize preparation time** (faster service, more customers)

But there's a problem: the most profitable drinks take longest to prepare! This is called a **trade-off**, and it's at the heart of real-world decision-making.

> **How to Use This Tutorial**
> Work through each section in order. Write code where marked "YOUR CODE BELOW" and verify with the provided assertions. This prepares you for the competition challenge!

Section 1: Visualizing the Trade-off

Let's start by importing our libraries and loading the product data:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from typing import List, Tuple, Dict

# Set random seed for reproducibility
np.random.seed(2025)
```

Product Data

Bean Counter's R&D team has developed potential new products. Each has been evaluated on multiple criteria:

```
# Product specifications
products = pd.DataFrame({
    'Product': ['Butterfly Lemonade', 'Ube Purple Latte', 'Tiger Nut Tea',
                'Saffron Rose Milk', 'Moringa Mint Cooler', 'Black Sesame
```

```
Frappé',
                'Sakura Cherry Coffee', 'Blue Spirulina Smoothie', 'Charcoal
Detox Shot',
                'Pandan Coconut Cream', 'Lavender Honey Foam', 'Dragon Fruit
Frappé',
                'Cascara Coffee Cherry', 'Nitro Matcha Float'],
    'Profit_Margin': [3.8, 4.1, 2.5, 4.3, 2.9, 3.5, 4.5, 3.2,
                      1.8, 3.6, 3.7, 4.8, 2.1, 3.4],  # € per unit
    'Prep_Time': [200, 240, 90, 270, 150, 210, 280, 180,
                  60, 220, 190, 300, 100, 140],  # seconds
    'Sustainability': [65, 40, 80, 55, 85, 62, 35, 60,
                       45, 68, 59, 50, 82, 78]  # score 0-100
})

print("Available Products:")
print(products.to_string(index=False))
```

```
Available Products:
               Product  Profit_Margin  Prep_Time  Sustainability
    Butterfly Lemonade            3.8        200              65
      Ube Purple Latte            4.1        240              40
         Tiger Nut Tea            2.5         90              80
     Saffron Rose Milk            4.3        270              55
   Moringa Mint Cooler            2.9        150              85
   Black Sesame Frappé            3.5        210              62
  Sakura Cherry Coffee            4.5        280              35
Blue Spirulina Smoothie           3.2        180              60
   Charcoal Detox Shot            1.8         60              45
  Pandan Coconut Cream            3.6        220              68
   Lavender Honey Foam            3.7        190              59
   Dragon Fruit Frappé            4.8        300              50
 Cascara Coffee Cherry            2.1        100              82
    Nitro Matcha Float            3.4        140              78
```

> **The Core Trade-off**
> Notice the fundamental conflict:
>
> - **Dragon Fruit Frappé**: Highest profit (€4.8) but slowest to make (300s!)
> - **Charcoal Detox Shot**: Fastest (60s = 1 minute) but lowest profit (€1.8)
>
> We can't have both maximum profit AND minimum prep time. We must choose a trade-off.

Exercise 1.1: Visualize the Profit-Speed Trade-off

Create a scatter plot to see this trade-off visually.

> **Creating a Scatter Plot**
>
> ```
> plt.figure(figsize=(10, 6))          # Create figure
> plt.scatter(x, y, s=size, alpha=0.7) # Plot points
> plt.annotate(label, (x, y), ...)     # Label each point
> plt.xlabel('X Label')                # Axis labels
> plt.ylabel('Y Label')
> plt.grid(True, alpha=0.3)            # Add grid
> ```

> **Before You Start**
>
> - X-axis: Profit Margin (want to maximize → move right)
> - Y-axis: Prep Time (want to minimize → move down)
> - **Best corner: Lower-right** (high profit, low time)
> - Use a for-loop to annotate all products

```
# YOUR CODE BELOW
# Create a scatter plot: Profit (x) vs Prep Time (y)
# Annotate each point with the product name
```

Section 2: Finding Non-Dominated Solutions

Not all products are worth considering. Some are **dominated**, so strictly worse than another option.

> **What is Dominance?**
> Product A **dominates** product B if:
>
> 1. A is **better or equal** on ALL objectives, AND
> 2. A is **strictly better** on AT LEAST ONE objective
>
> **Example:** If Product X has higher profit AND lower prep time than Product Y, then X dominates Y.
>
> You should NEVER choose a dominated product as there's always a better alternative!

Exercise 2.1: Implement Dominance Check

Write a function to check if a product is dominated by any other product.

> **Before You Start**
>
> - Compare the current product with ALL other products
> - For our objectives:
>   - Profit: higher is better (maximize)
>   - Prep Time: lower is better (minimize)
> - Return `True` if ANY product dominates the current one

```
def is_dominated(product_idx: int, products_df: pd.DataFrame) -> bool:
    """
    Check if a product at product_idx is dominated by any other product.

    Returns True if dominated, False otherwise.
    """
    current = products_df.iloc[product_idx]

    # YOUR CODE BELOW
    # Loop through all products and check if any dominates current
    # Dominance: other has >= profit AND <= prep_time, with at least one strict
< or >


    return False


# Test your function with the following
dominated = []
for i in range(len(products)):
    if is_dominated(i, products):
        dominated.append(products.iloc[i]['Product'])

print(f"Dominated products: {dominated}")


# Test your implementation
assert 'is_dominated' in dir(), "Define the is_dominated function"
assert len(dominated) >= 5, "Should find five dominated products"
assert 'Moringa Mint Cooler' in dominated, "Chai Tea should be dominated (lower
profit, slower than alternatives)"
print(f"Excellent! Found {len(dominated)} dominated products")
print("These should NEVER be chosen - better alternatives exist!")
```

Exercise 2.2: Find the Pareto Frontier

The **Pareto frontier** contains all non-dominated solutions - the only rational choices.

> **What is the Pareto Frontier?**
> The Pareto frontier (or Pareto set) is the set of ALL non-dominated solutions.
>
> **Why it matters:** These are the ONLY products worth considering. Every point on the frontier represents a different trade-off, and none is strictly better than the others.
>
> In the competition, you'll use this exact concept to find optimal fleet compositions!

> **Before You Start**
> - Create a boolean array `is_pareto` (all True initially)
> - For each product i, check if ANY product j dominates it
> - If dominated, set `is_pareto[i] = False`
> - Return only products where `is_pareto` is True

```
def find_pareto_frontier(products_df: pd.DataFrame) -> pd.DataFrame:
    """
    Find all non-dominated products (Pareto frontier).

    Returns DataFrame with only Pareto optimal products.
    """
    n = len(products_df)
    is_pareto = np.ones(n, dtype=bool)

    # YOUR CODE BELOW
    # For each product i, check if any product j dominates it


    return products_df[is_pareto]


# Find and display Pareto frontier based on your function
pareto_products = find_pareto_frontier(products)
print(f"\nPareto Frontier ({len(pareto_products)} products):")
print(pareto_products[['Product', 'Profit_Margin',
'Prep_Time']].to_string(index=False))


# Test your Pareto frontier function
assert 'find_pareto_frontier' in dir(), "Define find_pareto_frontier function"
assert len(pareto_products) >= 3, "Should find at least 3 Pareto optimal
products"
assert 'Saffron Rose Milk' in pareto_products['Product'].values, "Saffron Rose
Milk (€4.3 in 270s) should be on frontier"
print("Perfect! These are the ONLY products worth considering")
```

Visualizing the Pareto Frontier

Let's plot the Pareto frontier to see it clearly:

```
# Visualize Pareto frontier
plt.figure(figsize=(10, 6))

# Plot all products
plt.scatter(products['Profit_Margin'], products['Prep_Time'],
            alpha=0.3, color='gray', label='Dominated')

# Highlight Pareto frontier
```

```
pareto_idx = pareto_products.index
plt.scatter(products.loc[pareto_idx, 'Profit_Margin'],
           products.loc[pareto_idx, 'Prep_Time'],
           alpha=0.5, color='red',
           label='Pareto Frontier', zorder=5)

# Annotate Pareto products
for idx in pareto_idx:
    plt.annotate(products.loc[idx, 'Product'],
                 (products.loc[idx, 'Profit_Margin'], products.loc[idx,
'Prep_Time']),
                 fontsize=10, va='bottom')

plt.xlabel('Profit Margin (€)', fontsize=12)
plt.ylabel('Preparation Time (seconds)', fontsize=12)
plt.title('Pareto Frontier: Non-Dominated Products', fontsize=14,
fontweight='bold')
plt.legend(fontsize=11)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

> **Interpreting the Pareto Frontier**
> Each point on the frontier represents a different trade-off:
>
> - **Charcoal Detox Shot** (left): Fastest prep, lowest profit → "Speed Strategy"
> - **Dragon Fruit Frappé** (right): Highest profit, slowest prep → "Profit Strategy"
> - **Middle products**: Balanced compromises
>
> **None dominates the others** - your choice depends on your priorities!

Section 3: Making the Decision with Weighted Sums

How do we choose ONE product from the Pareto frontier? We use **weights** to express our priorities.

> **Why Normalization is Critical**
> Before combining objectives, we MUST normalize them:
>
> - Profit range: 1.8 to 4.8 (range = 3.0)
> - Prep Time range: 60 to 300 (range = 240)
>
> Without normalization, prep time dominates because 240 >> 3.0, even with equal weights!

Exercise 3.1: Normalize Objectives

Normalize both objectives to [0, 1] scale.

> **What is a pandas Series?**
> A **pandas Series** is like a single column from a DataFrame:
>
> `products['Profit_Margin']  # This is a Series (one column)`
>
> - Has built-in methods: `.min()`, `.max()`, `.mean()`
> - Supports arithmetic: `series - 5`, `series * 2`
> - Returns a new Series when you do math on it

> **Min-Max Normalization Formula**
>
> `normalized_value = (value - min_value) / (max_value - min_value)`
>
> - Minimum value → 0
> - Maximum value → 1
> - Preserves relative distances

```
def normalize_column(series: pd.Series) -> pd.Series:
    """Normalize a pandas Series to [0, 1] range."""
    # YOUR CODE BELOW
    # Hint: series.min() and series.max() give you the min/max values


# Normalize objectives with your new normalize_column function
products_norm = products.copy()
products_norm['Profit_Norm'] = normalize_column(products['Profit_Margin'])
products_norm['Prep_Norm'] = normalize_column(products['Prep_Time'])
print("Normalized values:")
print(products_norm[['Product', 'Profit_Norm',
'Prep_Norm']].to_string(index=False))


# Test normalization
assert 'normalize_column' in dir(), "Define normalize_column function"
assert products_norm['Profit_Norm'].min() >= 0, "Min should be >= 0"
assert products_norm['Profit_Norm'].max() <= 1, "Max should be <= 1"
assert abs(products_norm['Profit_Norm'].max() - 1.0) < 0.01, "Max should be
exactly 1.0"
print("Perfect! Objectives are now on the same scale [0, 1]")
```

Exercise 3.2: Calculate Weighted Score

Now combine the normalized objectives using weights.

> **Weighted Sum Method**
>
> `Score = w_profit × profit + w_speed × speed`
>
> Where weights sum to 1.0 (representing 100% of priorities).

> ⚠ **Warning**
> We want to **MINIMIZE** prep time, so use `(1 - prep_norm)` to flip it in the weighted score!

```
def calculate_score(profit_norm, prep_norm, w_profit, w_speed):
    """Calculate weighted score. Higher is better."""
    # YOUR CODE BELOW
    # Remember: minimize prep time means use (1 - prep_norm)

# Test different weight scenarios with your new function
scenarios = {
    'Profit Focus': {'w_profit': 0.7, 'w_speed': 0.3},
    'Balanced': {'w_profit': 0.5, 'w_speed': 0.5},
    'Speed Focus': {'w_profit': 0.3, 'w_speed': 0.7}
}

results = []
for name, weights in scenarios.items():
    products_norm['Score'] = calculate_score(
        products_norm['Profit_Norm'],
        products_norm['Prep_Norm'],
        weights['w_profit'],
        weights['w_speed']
    )
    best_idx = products_norm['Score'].idxmax()
    results.append({
        'Scenario': name,
        'Best Product': products.loc[best_idx, 'Product'],
        'Score': products_norm.loc[best_idx, 'Score']
    })

results_df = pd.DataFrame(results)
print("\nBest product for each scenario:")
print(results_df.to_string(index=False))

# Test weighted scoring
assert 'calculate_score' in dir(), "Define calculate_score function"
assert len(results_df) == 3, "Should have 3 scenarios"
assert all(results_df['Score'] > 0), "All scores should be positive"
assert all(results_df['Score'] <= 1), "All scores should be <= 1"
print("Excellent! Different weights lead to different optimal products")
```

Section 4: Hard Constraints

In reality, we often have **hard constraints**, requirements that MUST be met.

> **Constraints vs Objectives**
> - **Objective**: Something to optimize (minimize/maximize)
> - **Constraint**: A requirement that must be satisfied
>
> Example: Bean Counter wants sustainability ≥ 60

Exercise 4.1: Filter by Sustainability Constraint

Filter products to only those meeting the sustainability requirement.

```
# YOUR CODE BELOW
# Filter products where Sustainability >= 60
sustainability_threshold = 60
feasible_products = # YOUR CODE HERE

print(f"\nFeasible products (sustainability >= {sustainability_threshold}):")
print(feasible_products[['Product', 'Profit_Margin', 'Prep_Time',
'Sustainability']].to_string(index=False))


# Test constraint filtering
assert 'feasible_products' in dir(), "Create feasible_products variable"
assert len(feasible_products) >= 9, "Should have at least nine feasible
products"
assert all(feasible_products['Sustainability'] >= 60), "All products should
meet constraint"
print(f"✓ Great! {len(feasible_products)} products meet the sustainability
requirement")
```

Exercise 4.2: Pareto Frontier with Constraints

Find the Pareto frontier AMONG only the feasible products.

```
# YOUR CODE BELOW
# Apply find_pareto_frontier to feasible_products only
constrained_pareto = # YOUR CODE HERE

print(f"\nConstrained Pareto Frontier ({len(constrained_pareto)} products):")
print(constrained_pareto[['Product', 'Profit_Margin', 'Prep_Time',
'Sustainability']].to_string(index=False))


# Test constrained Pareto frontier
assert 'constrained_pareto' in dir(), "Create constrained_pareto variable"
assert len(constrained_pareto) <= len(feasible_products), "Pareto set should be
<= feasible set"
assert all(constrained_pareto['Sustainability'] >= 60), "All should meet
constraint"
print("Perfect! This is your constrained Pareto frontier")
```

```
print("These are the ONLY rational choices given the sustainability
requirement")
```

Visualizing Constraints and Pareto Frontier

```
# Visualize feasible region and Pareto frontier
plt.figure(figsize=(10, 6))

# Plot infeasible products
infeasible = products[products['Sustainability'] < 60]
if len(infeasible) > 0:
    plt.scatter(infeasible['Profit_Margin'], infeasible['Prep_Time'],
                alpha=0.3, color='red', marker='x', label='Infeasible')

# Plot feasible but dominated
feasible_dominated =
feasible_products[~feasible_products.index.isin(constrained_pareto.index)]
if len(feasible_dominated) > 0:
    plt.scatter(feasible_dominated['Profit_Margin'],
feasible_dominated['Prep_Time'],
                alpha=0.4, color='gray', label='Feasible (dominated)')

# Plot constrained Pareto frontier
plt.scatter(constrained_pareto['Profit_Margin'],
constrained_pareto['Prep_Time'],
        alpha=0.9, color='green',
           label='Constrained Pareto Frontier', zorder=5)

for idx in constrained_pareto.index:
    plt.annotate(products.loc[idx, 'Product'],
                 (products.loc[idx, 'Profit_Margin'], products.loc[idx,
'Prep_Time']),
                 fontsize=10, va='bottom')

plt.xlabel('Profit Margin (€)', fontsize=12)
plt.ylabel('Preparation Time (seconds)', fontsize=12)
plt.title('Constrained Pareto Frontier (Sustainability ≥ 60)', fontsize=14,
fontweight='bold')
plt.legend(fontsize=10)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

Exercise 4.3: Choose the Best Product with Weighted Scoring

Now that we have the constrained Pareto frontier, let's use weighted scoring to select the final product.

> **Before You Start**
> - Normalize the constrained Pareto products (not all products!)
> - Apply your `calculate_score` function
> - Find the product with the highest score
> - Use weights: 60% profit, 40% speed (w_profit=0.6, w_speed=0.4)

```
# YOUR CODE BELOW
# 1. Normalize the constrained Pareto products
constrained_norm = constrained_pareto.copy()
constrained_norm['Profit_Norm'] = # YOUR CODE HERE
constrained_norm['Prep_Norm'] = # YOUR CODE HERE

# 2. Calculate weighted scores (60% profit, 40% speed)
constrained_norm['Score'] = # YOUR CODE HERE

# 3. Find the best product
best_idx = # YOUR CODE HERE
best_product = products.loc[best_idx]

print(f"\nRECOMMENDED PRODUCT: {best_product['Product']}")
print(f"  Profit Margin: €{best_product['Profit_Margin']:.2f}")
print(f"  Prep Time: {best_product['Prep_Time']:.0f}s")
print(f"  Sustainability: {best_product['Sustainability']:.0f}")
print(f"  Weighted Score: {constrained_norm.loc[best_idx, 'Score']:.3f}")


# Test your final selection
assert 'constrained_norm' in dir(), "Create constrained_norm DataFrame"
assert 'Score' in constrained_norm.columns, "Calculate Score column"
assert 'best_idx' in dir(), "Find best_idx"
assert best_product['Sustainability'] >= 60, "Best product must meet
constraint"
print("Excellent! You've completed the full optimization workflow!")
```

Conclusion

Congratulations! You've mastered multi-objective optimization with constraints!

Key Takeaways

- **Trade-offs are inevitable** when optimizing multiple objectives
- **Dominated solutions** should never be chosen
- **Pareto frontier** contains all rational choices
- **Normalization** is critical before combining objectives
- **Weighted sums** let you express priorities
- **Hard constraints** limit the feasible region
- **Constrained Pareto frontier** = intersection of Pareto set and feasible region

What's Next?

In the competition, you'll apply these concepts to design EcoExpress's sustainable delivery fleet:

- **2 objectives**: Minimize total cost, Maximize service score
- **1 constraint**: CO2 emissions ≤ 111 g/km
- **Your task**: Generate fleet alternatives, find Pareto frontier, recommend the best one
- **Deliverable**: One-slide visualization showing your analysis

Good luck!

Bibliography